

Bounded Perturbation Transport and the Origin of Finite-Propagation Constraint

Continuation-Comparison Constraints on Standing-Relevant Transport

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Abstract

The preceding articles establish four dependencies: realized persistence is not reconstruction-null, standing-bearing persistence is boundedly localized, bounded loci induce continuation-neighborhood domains, and stable overlap among such domains induces an internal quotient comparison carrier. This paper derives the next physical role: the origin of finite-propagation constraint. It asks whether a standing-relevant perturbation burden can then be transported globally, pathlessly, or without admissible witnesses. It cannot.

The main result is a bounded perturbation transport theorem. In a fixed admissible realized persistence regime with local continuation domains and a quotient comparison carrier, standing-relevant perturbation burden can be transported from one domain to another only through admitted continuation-neighborhood paths equipped with boundary-preserving transport witnesses and quotient-stable comparison-class composition. Standing-relevant effects are therefore organized by admitted transport reach rather than asserted globally by default.

The theorem does not yet derive a numerical speed, temporal parameter, light cone, Lorentzian causal structure, field equation, signal dynamics, hyperbolic propagation law, local algebra, or empirical causality claim. The physical-role claim is nonetheless substantive: finite-propagation constraint first enters as bounded, witnessed perturbation transport. Standard finite-speed, causal, and field-propagation formalisms are downstream specializations of this transport role.

1 Introduction

1.1 Theorem claim and physical role

This article derives the propagation-facing role before numerical speed or causal geometry is available. The theorem claim is bounded perturbation transport:

$$(\text{LCD}_D, C_D) \wedge \text{Perturb}_D \wedge \text{TransportAdm}_D \Rightarrow \text{BoundTrans}_D.$$

The physical-role claim is that finite-propagation constraint first appears as the requirement that standing-relevant perturbation burden cannot be global, pathless, comparison-free, or identity-resetting. It must be carried by admitted continuation-neighborhood paths with boundary-preserving and quotient-stable witnesses.

Thus the result is not a denial of propagation physics. It is the structural origin of the finite-propagation role that later time, metric, field-equation, and causal-order formalisms specialize.

The preceding articles establish a cumulative structural ladder. Minimal non-vacuity gives the persistence floor: a target typed as carrying realized persistence inside a fixed admissible

continuation regime cannot remain null with respect to its own persistence-sensitive reconstruction structure [1]. In schematic form,

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

Standing localization then shows that non-vacuous standing-bearing persistence cannot remain global, diffuse, trace-only, hidden, or unbounded once perturbation and restorative continuation constraints are admitted [2]:

$$\text{NonVacRec}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv_B} \neq \emptyset.$$

Continuation-neighborhood induction shows that bounded standing-localization classes cannot support cross-locus continuation without boundary-compatible adjacency, standing-verdict separability, and local continuation domains [3]:

$$[B_D]_{\equiv_B} \wedge \text{BoundaryFix}_D \wedge \text{ContCompat}_D \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{Sep}_D, \text{LCD}_D).$$

Proto-metric carrier induction then shows that stable overlap and standing-preserving comparison among local continuation domains require an internal quotient comparison carrier [4]:

$$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \wedge \text{OverlapStab}_D \wedge \text{TransportCoherence}_D \wedge \text{QuotStab}_D \Rightarrow C_D.$$

The present analysis begins at this point. Local continuation domains and quotient comparison carriers are available, but propagation structure is not. There is still no time parameter, signal velocity, causal cone, field equation, local algebra net, metric tensor, or spacetime path. The resulting question is narrow:

Once local continuation domains and quotient comparison carriers exist, can standing-relevant perturbation burden be transported without bounded, witnessed path structure?

The answer is no. If a perturbation affects every domain by default, standing-verdict separability collapses. If it passes from one local domain to another without an admitted continuation-neighborhood path, the transfer imports a hidden selector, external channel, identity reset, domain drift, or AMetric default. If it traverses a path without quotient-stable transport witnesses, comparison-class coherence is lost. Perturbation transport is therefore bounded before any later theory can interpret such boundedness as propagation.

1.2 Main result

The governing sentence is:

Standing-relevant perturbation burden can be transported only through admitted continuation-neighborhood paths equipped with boundary-preserving transport witnesses and quotient-stable comparison-class composition.

In compact form,

$$(\text{LCD}_D, C_D) \wedge \text{Perturb}_D \wedge \text{TransportAdm}_D \Rightarrow \text{BoundTrans}_D.$$

Here BoundTrans_D denotes bounded perturbation transport. A secondary consequence is transport-bounded organization:

$$\text{BoundTrans}_D \Rightarrow \text{ProtoCausal}_D,$$

where ProtoCausal_D is used only in the restricted sense defined below: standing-relevant perturbation effects are constrained by admitted transport paths rather than being globally or pathlessly available. Neither expression denotes finite physical speed, temporal causality, relativistic causal order, microcausality, or a field propagation law.

1.3 Scope

The analysis establishes five claims.

- (i) **Global perturbation by default is inadmissible.** A standing-relevant perturbation that affects all local continuation domains without admitted transport structure collapses local burden assignment and standing-verdict separability.
- (ii) **Pathless transfer is inadmissible.** Cross-domain perturbation transfer without an admitted continuation-neighborhood path imports hidden selection, external channeling, identity reset, or domain drift.
- (iii) **Transport requires witnesses.** Standing-relevant burden can be transported only when boundary-compatible witnesses preserve source burden, target burden, quotient identity, restoration status, local-domain role, and comparison class.
- (iv) **Transport composition must be quotient-stable.** Along admissible paths, transport witness classes compose only where intermediate boundary traces and comparison classes remain coherent.
- (v) **Transport-bounded organization is only a precondition.** The result supplies the bounded path-mediated structure required before finite propagation or causal ordering can be formulated.

The paper does not yet specialize bounded transport into a numerical speed. Speed requires a metric scale, temporal parameter, unit convention, and comparison between path extent and temporal duration. Lorentzian cones, causal order in spacetime, hyperbolic partial differential equations, field propagation, algebraic microcausality, and empirical finite-signal constraints are later strengthening layers. The positive result is the finite-propagation role at its earliest layer: standing-relevant effect must have admitted, bounded, witness-mediated transport reach.

1.4 Relation to standard propagation and causal frameworks

Standard finite propagation results normally appear inside already-specified analytic or geometric structures: hyperbolic partial differential equations on domains, causal cones in Lorentzian manifolds, or local algebra assignments over spacetime regions [6, 7, 8, 9, 10, 11, 12]. In those settings, locality, regions, time, and propagation equations are already available. The present analysis targets an earlier layer. It asks why a perturbation cannot acquire standing-relevant reach globally or pathlessly once the preceding stages have fixed localization, continuation-neighborhoods, and comparison carriers.

The result is therefore not a replacement for causal geometry or hyperbolic evolution theory. It is a precondition result: any later finite-speed theory must respect the bounded transport organization already required by standing-bearing persistence. Conversely, a theory that permits arbitrary pathless standing-relevant perturbation transfer would erase the localization, separability, and comparison structure established by the preceding stages.

2 Transport strengthening hierarchy

The language of propagation covers several layers that must be kept distinct. The present paper proves only the bounded-transport and transport-bounded organization layers.

Level	Structure
T0	Bounded standing-localization: perturbation burden is assignable to bounded standing-bearing loci.
T1	Continuation-neighborhood mediation: cross-domain burden transfer proceeds through admitted adjacency paths.
T2	Witness-mediated perturbation transport: source burden, target burden, boundary traces, quotient identity, restoration status, and comparison class are preserved.
T3	Quotient-stable transport composition: transport witness classes compose under $\odot_D$ where boundary traces are composable.
T4	Transport-bounded organization: standing-relevant effects are organized by admitted transport paths rather than global availability.
T5	Finite physical speed: metric scale, temporal parameterization, units, and a propagation law are available.
T6	Relativistic causal structure: Lorentzian cone structure, temporal orientation, and field or algebraic dynamics are available.

The theorem below establishes T2–T4, conditional on T0–T1 and the quotient comparison carrier supplied by the preceding stage. It does not establish T5 or T6. This hierarchy fixes the scope of the bounded transport theorem and prevents it from being mistaken for a finite velocity theorem.

Remark 2.1 (No finite speed yet). Finite speed requires at least a metric or scale structure, a temporal parameter, comparable units of duration, field or signal dynamics, and a rule comparing path extent with temporal interval. None of these are introduced here. Bounded perturbation transport is the structural condition without which a later finite-propagation theorem would have no admissible perturbation-transport object to constrain.

3 Dependency order and definitions

The argument is conditional on the earlier layers. It does not re-prove non-vacuity, localization, continuation-neighborhoods, or comparison-carrier induction.

Prior layer	Imported role
Minimal non-vacuity	Perturbation cannot act through null realized structure.
Standing localization	Perturbation burden requires bounded standing-localization and restorative evaluability.
Continuation-neighborhood induction	Cross-domain transport must traverse admitted continuation-neighborhood paths.
Proto-metric carrier induction	Transport comparison requires quotient comparison carrier C_D and partial composition $\odot_D$.
Boundary-trace fixation	Transport transitions require boundary-compatible witnesses, not default representatives.
AMetric boundary	No primitive global jump, hidden transmission channel, default path selection, or instantaneous all-to-all transfer may be imported.
Scale-transport discipline	Standing-bearing transport must preserve quotient behavior under lawful redescription and regime transition.
Entanglement boundary discipline	Path-constrained perturbation transport is not Bell-local hidden-variable doctrine.

Definition 3.1 (Fixed admissible realized persistence regime). A fixed admissible realized persistence regime D is a same-domain continuation setting in which standing-bearing persistence, perturbation burden, restoration status, and transport are evaluated without hidden selector import, fail-open relabeling, cross-domain laundering, or identity reset. All necessity claims in this paper are structural and conditional on this fixed-domain setting.

Definition 3.2 (Local continuation domain). For a bounded standing-localization class L , the local continuation domain is

$$\text{LCD}_D(L) = \{K : L \rightsquigarrow_{\text{Adj}_D}^* K\},$$

where $L \rightsquigarrow_{\text{Adj}_D}^* K$ denotes an admissible continuation-neighborhood path generated by boundary-compatible adjacency. A local continuation domain is not a spatial region. It is the domain generated by admitted continuation paths preserving boundary compatibility, quotient behavior, target identity, restoration status, and fail-closed standing verdicts.

Definition 3.3 (Quotient comparison carrier). The quotient comparison carrier C_D is the internal carrier of stable comparison witnesses among local continuation domains, constructed as comparison-witness equivalence classes under standing-bearing quotient preservation. It is imported from the proto-metric carrier layer. It need not be numerical, total, symmetric, scalar-valued, linearly ordered, continuous, additive, or globally comparable.

Definition 3.4 (Perturbation). A perturbation $p \in \text{Pert}_D(U)$ in a local continuation domain U is standing-relevant iff it changes, threatens, conditions, or tests at least one of the following: standing verdict, restoration status, boundary-trace compatibility, quotient identity, comparison class in C_D , or local-domain role. A perturbation is not yet an external force, field excitation, signal, particle interaction, or physical cause.

Definition 3.5 (Perturbation burden). The perturbation burden of p in U , written $\text{Bur}_D(p, U)$, is the continuation-sensitive consequence assignment induced by p relative to standing retention and restorative continuation constraints. Burden is not a scalar magnitude unless a later strengthening supplies a scalar representation. It is the role that must remain assignable, transportable, or fail-closed if the perturbation is to be standing-relevant.

Definition 3.6 (Transport witness). A transport witness

$$\tau_D(U, V, p) : (U, p) \rightsquigarrow (V, p')$$

is admissible only when it preserves all of the following roles:

- (a) source perturbation burden;
- (b) target perturbation burden;
- (c) boundary-trace compatibility;
- (d) quotient identity;
- (e) restoration status;
- (f) comparison class in C_D ;
- (g) local-domain role.

A claimed transfer without such a witness is not admissible perturbation transport internal to D . It is hidden/default transfer, identity reset, external-channel import, domain drift, or notation-level routing.

Definition 3.7 (Bounded perturbation transport). $\text{BoundTrans}_D(p, U, V)$ holds iff there exists an admissible finite or schematically finite continuation-neighborhood path

$$U = U_0 \text{Adj}_D U_1 \text{Adj}_D \cdots \text{Adj}_D U_n = V$$

carrying p through transport witnesses $\tau_{i,i+1}$ such that each transition is boundary-compatible, restoration-evaluable, quotient-stable, and comparison-class composable under $\odot_D$ where composition is defined. Bounded transport excludes pathless global transfer, hidden selector import, cross-domain drift, identity reset, skin-only transfer, and primitive metric routing.

Definition 3.8 (Proto-causal organization). ProtoCausal_D means only the partial organization of standing-relevant perturbation effects by admitted transport paths, transport witnesses, and quotient-stable comparison classes. It does not mean temporal causation, event chronology, finite velocity, light-cone structure, Lorentzian causality, signal propagation, field-theoretic microcausality, partial differential equation propagation, or empirical causal dynamics.

4 No primitive transport import

A transport route cannot be introduced before the relevant standing, boundary, and comparison roles have been fixed. Otherwise the route acts as a selector rather than as admissible continuation structure.

Theorem 4.1 (No primitive transport import). *In a fixed admissible realized persistence regime, a perturbation transport route cannot be introduced as primitive if it selects target domains, path order, reach, or comparison class independently of boundary-compatible continuation witnesses. Such a route is not transport of standing-relevant burden; it is external selection or notation-level routing.*

Proof. A primitive route fixes where the perturbation burden goes before the burden-preserving continuation relation has been witnessed. If it chooses a target, it selects a domain without boundary trace. If it chooses a path, it imports route structure independently of continuation-neighborhood mediation. If it chooses reach, it fixes effect before transport. If it chooses comparison class, it imports the quotient comparison role rather than deriving it from admissible witnesses. In each case the route performs the role of a hidden selector, AMetric default, or skin-level diagram. It therefore cannot function as fixed-domain standing-preserving perturbation transport. $\square$

This theorem does not forbid primitive transport structures in an external formal model. It says only that such structures cannot discharge the internal role of admissible perturbation transport within the realized persistence regime unless they are themselves fixed by boundary-compatible continuation.

5 Global perturbation collapse

A perturbation can be standing-relevant only if its burden is assignable. Assignment does not require a metric location, coordinate chart, or spatial point. It requires a bounded continuation-sensitive domain in which the perturbation's effect can be evaluated as standing-relevant or fail-closed. If the perturbation is instead allowed to affect all local continuation domains by default, the standing-localization and separability results established above are erased.

Lemma 5.1 (Global perturbation collapse). *Let D be a fixed admissible realized persistence regime with standing-localization classes, continuation-neighborhood domains, and standing-verdict separability. If a perturbation p is allowed to standing-relevantly affect every local continuation domain without an admitted transport path, boundary witness, or comparison-class composition, then local burden assignment and standing-verdict separability collapse.*

Proof. By standing localization, perturbation burden is evaluated over bounded standing-localization classes. By continuation-neighborhood induction, standing verdicts in distinct local continuation domains remain independently evaluable except through admitted adjacency paths. Suppose p affects every domain without admitted transport. Then the effect of p is not assigned through any bounded source, admitted boundary transition, quotient comparison class, or local continuation path. For any domains U, V , failure, restoration, or standing alteration in U becomes automatically standing-relevant to V . This is indistinguishable from global standing collapse: separability no longer marks independent evaluability, and bounded localization no longer constrains burden assignment. The perturbation is therefore not compatible with fixed-domain standing-bearing persistence. $\square$

The point is not that global structures are forbidden in all senses. A regime may possess global constraints, conservation laws, or boundary-wide compatibility conditions when those are admissibly fixed. The excluded structure is default global standing-relevant perturbation effect without a boundary witness or transport organization. If a global relation is witnessed, it is not pathless by default; it is an admitted global boundary relation. If it is not witnessed, it is an AMetric default.

6 Pathless transport failure

A second escape route is pathless transfer. One might grant local burden assignment but allow perturbation burden to move from a source domain U to a target domain V without traversing

admitted continuation-neighborhood structure. That move also fails.

Lemma 6.1 (Pathless transport failure). *If standing-relevant perturbation burden is transferred from U to V without an admitted continuation-neighborhood path or transport witness, then the transfer is either hidden selector import, external channel import, identity reset, cross-domain drift, or AMetric default selection. None is admissible as fixed-domain perturbation transport.*

Proof. A cross-domain burden transfer must explain how the source burden remains the same standing-relevant role at the target. Without an admitted path, no intermediate boundary traces are fixed. Without a witness, no quotient identity or restoration status is preserved. If the target is chosen nevertheless, a selector has been imported. If the transfer occurs through an untyped channel, an external relation has been imported. If the target burden is simply declared, identity has reset. If source and target are evaluated under different regime standards, the transfer drifts domains. If none of these is supplied, the result is a default AMetric selection. Each option violates fixed-domain admissibility. $\square$

This lemma does not say that all admissible transport must be visually local, graph-local, or metric-local. It says only that transport must be witnessed by the regime’s own boundary and comparison structure. A global boundary witness, if admitted, is a witness; it is not pathless transfer by default.

7 Transport witnesses and boundary preservation

The transport witness is the central object of the paper. It is what prevents perturbation transfer from degenerating into hidden routing or re-labeling.

Proposition 7.1 (Transport witness requirement). *If a standing-relevant perturbation burden is admissibly transported from U to V , then there exists an admissible transport witness preserving source burden, target burden, boundary-trace compatibility, quotient identity, restoration status, comparison class in C_D , and local-domain role:*

$$\text{TransportAdm}_D(p, U, V) \Rightarrow \exists \tau_D(U, V, p).$$

Proof. Admissible transport is not merely the appearance of an effect in V . It must be transport of the standing-relevant burden originating in U . Source and target burdens must therefore be linked. Because the transport crosses local continuation domains, boundary traces must be preserved. Because the burden remains the same standing-bearing role, quotient identity and restoration status must be preserved. Because Paper 4 supplies comparison classes as the internal comparison carrier, the transport must preserve the comparison class that types the burden’s continuation role. Finally, because the target is a local continuation domain, the local-domain role must remain fixed. These preservation requirements are precisely the admissible transport witness. Without such a witness the claimed transport is hidden/default transfer, identity reset, external-channel import, or domain drift. $\square$

The proposition is deliberately role-level. It does not require a scalar magnitude, metric path, temporal interval, or signal model. It requires only the boundary and quotient data needed for standing-bearing perturbation burden to remain the same burden under transport.

8 Quotient-stable transport composition

Paper 4 introduced partial composition $\odot_D$ for quotient comparison classes. The same discipline applies to perturbation transport. A path may carry a burden through multiple local continuation domains only when the relevant witness classes compose. This does not assume global path independence.

Proposition 8.1 (Quotient-stable transport composition). *For an admitted continuation-neighborhood path*

$$U_0 \text{Adj}_D U_1 \text{Adj}_D \cdots \text{Adj}_D U_n,$$

transport of a standing-relevant perturbation burden is admissible only where the witness classes

$$[\tau_{01}], [\tau_{12}], \dots, [\tau_{n-1,n}]$$

are composable under $\odot_D$. If a direct witness $[\tau_{0n}]$ and the composed path claim the same standing-bearing perturbation role, then

$$[\tau_{0n}] = [\tau_{01}] \odot_D [\tau_{12}] \odot_D \cdots \odot_D [\tau_{n-1,n}].$$

No global path independence is assumed; equality is required only when the regime asserts equivalence of the direct and composed transport roles.

Proof. Each adjacent transport step preserves source burden, target burden, boundary trace, quotient identity, restoration status, comparison class, and local-domain role. Such preservation is meaningful across a path only if the target boundary of one step matches the source boundary of the next and the corresponding comparison classes remain coherent. Where these conditions hold, $\odot_D$ is defined. If a direct witness and a composed path claim the same perturbation role, quotient stability requires them to determine the same witness class. Otherwise the same burden would acquire two inequivalent standing-bearing transport roles inside the same fixed regime. $\square$

This condition is weaker than conservative dynamics, metric path-independence, or action minimization. It states only that same-role transport must remain quotient-coherent.

9 Bounded perturbation transport

The preceding lemmas yield the central theorem.

Theorem 9.1 (Bounded perturbation transport). *Let D be a fixed admissible realized persistence regime with local continuation domains and an internal quotient comparison carrier C_D . Any standing-relevant perturbation burden transported from U to V without identity reset, hidden selector import, external-channel import, or domain drift must be transported through admitted continuation-neighborhood paths equipped with boundary-preserving transport witnesses and quotient-stable comparison-class composition:*

$$(\text{LCD}_D, C_D) \wedge \text{Perturb}_D \wedge \text{TransportAdm}_D \Rightarrow \text{BoundTrans}_D.$$

Proof. Let $p \in \text{Pert}_D(U)$ be standing-relevant and suppose its burden is transported to V . By standing localization, the source burden is assignable only over bounded standing-bearing structure. By continuation-neighborhood induction, cross-domain relations among bounded loci are mediated by admitted adjacency paths. If no such path is supplied, pathless transport failure applies. If the

effect is declared across all domains, global perturbation collapse applies. If a path is supplied but no transport witnesses are supplied, the transport witness requirement fails. If witnesses are supplied but fail to compose where the regime asserts same-role transport, quotient-stable transport composition fails. Hence every admissible same-domain transport must proceed through admitted continuation-neighborhood paths preserving boundary traces, quotient identity, restoration status, comparison class, and local-domain role. That is bounded perturbation transport. $\square$

The theorem does not assert finite velocity, temporal sequence, or causal dynamics. It establishes the admissible transport object that later finite-propagation or causal-ordering theories would need to refine.

10 Transport-bounded organization

A bounded transport theorem has a propagation-facing consequence, but the consequence must be named carefully. It is not causality. It is the partial organization of standing-relevant perturbation effects by admitted transport paths.

Corollary 10.1 (Transport-bounded organization). *If BoundTrans_D holds, then standing-relevant perturbation effects are organized by admitted transport paths, boundary witnesses, and quotient-stable comparison classes. In the restricted terminology of this paper,*

$$\text{BoundTrans}_D \Rightarrow \text{ProtoCausal}_D.$$

Proof. By bounded transport, a perturbation burden from U can affect V only through an admitted path with transport witnesses. Thus the effect is not global by default and not pathless. The admitted path and witness composition partially organize which target domains can receive the burden and under which quotient roles. This is exactly ProtoCausal_D in the restricted definition above. $\square$

The term “proto-causal” is used only to mark the absence of default global effect. It does not introduce time, event causation, signal dynamics, or relativistic causal order.

11 Transport reach and no pathless global effect

Bounded transport determines an internal object: the reach set of a perturbation.

Definition 11.1 (Transport reach). For a perturbation p in a local continuation domain U , define

$$\text{Reach}_D(U, p) = \{V : \text{BoundTrans}_D(p, U, V)\}.$$

This is the set of local continuation domains reachable from U by admitted perturbation-transport witnesses for p .

Remark 11.2 (What reach is not). $\text{Reach}_D(U, p)$ is not a light cone, causal future, temporal future, signal cone, metric ball, field-support region, or spacetime domain of dependence. It is a domain-internal reach set under admitted perturbation transport. It may be partial, path-dependent, witness-dependent, and non-symmetric.

Corollary 11.3 (No pathless global perturbation effect). *A standing-relevant perturbation cannot affect arbitrary local continuation domains unless admitted transport witnesses or admitted global boundary witnesses exist. If such witnesses exist, the effect is witnessed rather than pathless; if no such witnesses exist, the claimed effect is inadmissible as standing-relevant perturbation transport.*

Proof. If arbitrary targets are affected without witnesses, global perturbation collapse applies. If an admitted transport path exists, the effect is bounded transport. If an admitted global boundary witness exists, then the effect is globally witnessed rather than pathless. In the remaining case no admissible witness or path fixes the standing-relevant transfer, so the claimed effect is a hidden/default transfer or AMetric selection. $\square$

This corollary is not a physical speed limit; it is a no-default-global-effect result.

12 Relation to entanglement and Bell locality

The result is not a local hidden-variable doctrine. It does not assert Bell locality, factorization of joint probabilities, separability of quantum states, absence of entanglement, or absence of nonclassical correlations [13, 14]. Nor does it deny globally witnessed compatibility structures. The claim is narrower: standing-relevant perturbation burden does not transfer by default across realized continuation domains without admitted transport structure.

This distinction follows from the earlier treatment of nonclassical compatibility as boundary-level structure rather than as ordinary local mechanism [5]. A boundary-wide compatibility relation, if admitted, is a witness. It is not an unwitnessed global perturbation effect. Conversely, Bell-nonlocal correlation structure does not license pathless standing-burden transport inside a fixed realized persistence regime. The two issues are typed differently.

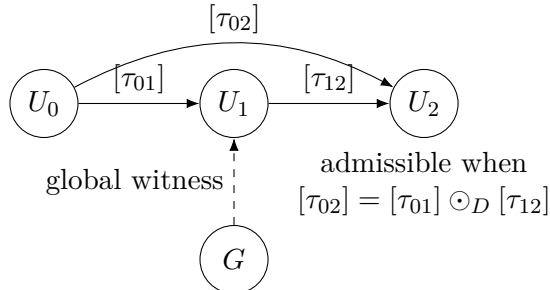
13 Finite witness model

The finite model is a witness model only; it instantiates one admissible realization of comparison-constrained transport and does not define the general transport structure.

Let

$$G_D = (V, E, \tau, C_D)$$

where vertices V are local continuation domains, edges E are admitted adjacency transitions, τ_e are perturbation transport witnesses, C_D supplies quotient comparison classes, and $\odot_D$ composes witness classes only where boundary traces match.



The model separates five cases.

- (i) **Complete global effect without edges:** every vertex is affected without witnesses; standing-verdict separability collapses.
- (ii) **Hidden edge:** a target is selected without an admitted boundary transition; selector import occurs.
- (iii) **Arbitrary transport label:** an edge receives a comparison class not determined by witnesses; comparison import occurs.
- (iv) **Path-incoherent transport:** a direct witness and composed witnesses claim the same perturbation role but yield inequivalent classes; quotient instability occurs.
- (v) **Composable admitted witnesses:** boundary traces match and witness classes compose under $\odot_D$; bounded transport is realized.

A global node G in the diagram is admissible only when it represents an actual global boundary witness. Without such a witness it would be merely a hidden route.

14 Route exhaustion

The possible candidates for standing-relevant perturbation transport are exhausted by broad failure classes.

Theorem 14.1 (Perturbation transport route exhaustion). *Any candidate perturbation transport structure for a fixed admissible realized persistence regime is one of the following:*

- (i) *admitted bounded transport;*
- (ii) *global or pathless transfer;*
- (iii) *hidden/default transfer;*
- (iv) *identity-reset transfer;*
- (v) *path-incoherent transfer;*
- (vi) *cross-domain transfer;*
- (vii) *skin-only or formal-only transfer.*

Classes (ii)–(vii) fail as standing-relevant fixed-domain perturbation transport. Only class (i) survives.

Proof. If transport is mediated by admitted continuation-neighborhood paths, boundary-compatible witnesses, and quotient-stable comparison classes, it is class (i). If not, then either no path is supplied, giving global or pathless transfer; an unwitnessed selector, default, or channel is supplied, giving hidden/default transfer; the target burden is declared anew, giving identity reset; the route changes comparison class for the same role, giving path incoherence; source and target are evaluated under different domain standards, giving cross-domain transfer; or the transfer is only diagrammatic, giving skin-only transfer. Global/pathless transfer collapses separability, hidden/default transfer violates AMetric boundary discipline, identity reset fails persistence, path incoherence fails quotient stability, cross-domain transfer violates fixed-domain admissibility, and skin-only transfer carries no standing-bearing role. Hence only admitted bounded transport survives. $\square$

This exhaustion is conditional on the target being standing-relevant perturbation transport. It does not classify all mathematical relations, nonclassical compatibilities, global constraints, or external descriptions. It classifies the options available when a perturbation’s burden is to be transported while preserving the preceding structural layers.

15 Relation to finite propagation and causal structure

The present result supplies three ingredients that later finite-propagation and causal-structure papers can use:

- (i) **bounded transport:** standing-relevant perturbation effects must be mediated by admitted continuation-neighborhood paths;
- (ii) **reach organization:** perturbation effects have a domain-internal reach set generated by transport witnesses;
- (iii) **transport-bounded precedence:** if p ’s burden from U affects V , the effect is organized by the existence of a transport path and its witness composition.

These structures are necessary for finite-speed propagation but not sufficient for it. Finite speed requires additional metric and temporal strengthening. Relativistic causal structure requires Lorentzian signature, cone structure, and dynamical interpretation. Field propagation requires carrier equations or local algebraic support. The present paper deliberately stops before those layers.

The phrase “proto-causal” is used in a restricted sense: it records path-mediated standing-relevant effect constraints, not temporal causation in a spacetime manifold. This restriction is essential. If temporal ordering were imported at this stage, it would bypass the later analysis of irreversible continuation ordering. If metric cones were imported here, they would bypass the metric and propagation-strengthening layers. The present analysis instead establishes the bounded-transport organization that those later layers must preserve.

16 Normal form

The result can be stated in normal form:

$$\boxed{(\text{LCD}_D, C_D) \wedge \text{Perturb}_D \wedge \text{TransportAdm}_D \Rightarrow \text{BoundTrans}_D \Rightarrow \text{ProtoCausal}_D}$$

where BoundTrans_D is bounded, witness-mediated perturbation transport and ProtoCausal_D is transport-bounded organization in the restricted sense above. Neither term denotes finite physical speed. The normal form emphasizes dependency order: local continuation domains and quotient comparison carriers must already be available before perturbation effects can be organized by bounded transport.

17 Conclusion

Paper 1 established non-vacuous realized persistence. Paper 2 established bounded standing-localization. Paper 3 established continuation-neighborhood domains. Paper 4 established quotient-stable comparison carriers. This paper derives the next physical role: bounded perturbation transport, the structural origin of finite-propagation constraint.

Standing-relevant perturbation transport cannot be global, pathless, comparison-free, or identity-resetting while preserving the earlier layers. It must proceed through admitted continuation-neighborhood paths whose transport witnesses preserve boundary traces, quotient identity, restoration status, local-domain role, perturbation burden, and comparison-class composition. Standing-relevant effects are therefore organized by admitted transport reach rather than asserted globally by default.

No numerical speed, time parameter, light cone, field equation, local algebra, temporal order, or relativistic causal structure has been derived. The physical interpretation is positive: the paper derives the finite-propagation role before its standard formal species. Later causal, relativistic, hyperbolic, algebraic, or field-theoretic propagation structures must specialize this bounded transport role rather than bypass it.

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